

MATH 302: Homework 1

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Question 1

Question. Show that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Definition (1.1 Metric Space Properties). A set is a metric space if it is equipped with a function $d : V \times V \rightarrow \mathbb{R}$ such that for all $x, y, z \in V$:

Property	Description
I. Identity of Indiscernibles	$d(x, y) = 0 \Leftrightarrow x = y$
II. Symmetry	$d(x, y) = d(y, x)$
III. Triangle Inequality	$d(x, y) \leq d(x, z) + d(z, y)$
IV. Non-Negativity	$d(x, y) \geq 0$

Let there be a normed vector space $(V, \|\cdot\|)$. Define $d(x, y) = \|x - y\|$. We will show that d satisfies the four properties of a metric space.

Identity of Indiscernibles

Observe that the “Identity of Indiscernibles” property states that $d(x, y) = 0$ if and only if $x = y$. In other words given $x, y \in V$ and the distance between them is zero, then they must be the same point.

By the definition of d , we have:

$$d(x, y) = \|x - y\|$$

By the definition of a norm, $\|v\| = 0$ if and only if $v = 0$. Applying this to $v = x - y$:

$$d(x, y) = 0 \Leftrightarrow \|x - y\| = 0 \Leftrightarrow x - y = 0 \Leftrightarrow x = y,$$

where the final step follows by adding y to both sides. Thus, the “Identity of Indiscernibles” property holds. ■

Symmetry

Observe the “Symmetry” property holds if the distance from x to y is the same as the distance from y to x . In other words, given $x, y \in V$, we have that $d(x, y) = d(y, x)$.

We can test this assertion by substituting the definition of d into the equation:

$$d(x, y) = \|x - y\| = \|-(y - x)\| = \|y - x\| = d(y, x)$$

Thus, the “Symmetry” property holds. ■

Triangle Inequality

Observe the “Triangle Inequality” property states the distance between two points x and y is less than or equal to the sum of the distances from x to a third point z and from z to y . In other words, given $x, y, z \in V$, we have that $d(x, y) \leq d(x, z) + d(z, y)$.

We can test this assertion by substituting the definition of d into the equation:

$$\begin{aligned}d(x, z) &= \|x - z\| \wedge d(z, y) = \|z - y\| \wedge d(x, y) = \|x - y\| \\d(x, z) + d(z, y) &= \|x - z\| + \|z - y\|\end{aligned}$$

Observe the following:

$$\|x - y\| = \|(x - z) + (z - y)\| \leq \|x - z\| + \|z - y\|$$

Thus, the “Triangle Inequality” property holds. ■

Non-Negativity

Observe the “Non-Negativity” property states that the distance between two points x and y is always greater than or equal to zero. In other words, there can never be a negative distance between two points.

As the distance function d is defined as the norm of the difference between two points, we have that $d(x, y) = \|x - y\|$. By the definition of a norm, we know that $\|v\| \geq 0$ for all $v \in V$. Therefore, $d(x, y) \geq 0$ for all $x, y \in V$. ■

Conclusion

As we have shown that d satisfies all four properties of a metric space, we conclude that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Question 2

Question. On $\mathbb{R}^n$ define, for any $x = (x_1, \dots, x_n)$,

$$\|x\|_1 = \sum_{j=1}^n |x_j| \quad \text{and} \quad \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}.$$

Show that both $\|x\|_1$ and $\|x\|_\infty$ are norms on $\mathbb{R}^n$.

Definition (Definition of a Norm) . An element is a norm if the following is met:

Property	Description
I. Positive Definite	$\ x\ = 0 \iff x = 0$
II. Absolute Homogeneity	$\ ax\ = \ a\ \ x\ $
III. Triangle Inequality	$\ x + y\ \leq \ x\ + \ y\ $

WWTS $\|x\|_1$ and $\|x\|_\infty$ maintain these properties on the vector space $\mathbb{R}^n$.

Positive Definite

Let $p := (\|x\| = 0)$ and $q := (x = 0)$. WWTS $p \iff q$ for both $\|\cdot\|_1$ and $\|\cdot\|_\infty$.

Forward Proof

WWTS $p \implies q$. Equivalently, we prove the contrapositive $\neg q \implies \neg p$, that is

$$(x \neq 0) \implies (\|x\| \neq 0).$$

Suppose $x = (x_1, \dots, x_n) \neq 0$. Then x differs from the zero vector in at least one coordinate, so there exists an index $a \in \{1, \dots, n\}$ with $x_a \neq 0$. By the properties of absolute value,

$$|x_a| > 0 \quad \text{and} \quad |x_j| \geq 0 \quad \forall j \in \{1, \dots, n\}.$$

Case $\|x\|_1$

Splitting the sum at the index a and bounding each piece below:

$$\begin{aligned} 0 &< |x_a| \\ + 0 &\leq \sum_{j \neq a} |x_j| \\ \hline 0 &< \sum_{j=1}^n |x_j| = \|x\|_1 \end{aligned}$$

Hence $\|x\|_1 > 0$, and in particular $\|x\|_1 \neq 0$. ■

Case $\|x\|_\infty$

Since $|x_a| \in \{|x_1|, \dots, |x_n|\}$ and the maximum of a set is an upper bound for every element of that set,

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} \geq |x_a| > 0.$$

Hence $\|x\|_\infty > 0$, and in particular $\|x\|_\infty \neq 0$. ■

Back Proof

WWTS $q \implies p$. Suppose $x = 0$, so that $x_j = 0$ for every $j \in \{1, \dots, n\}$, and therefore $|x_j| = 0$ for every such j .

Case $\|x\|_1$

$$\|x\|_1 = \sum_{j=1}^n |x_j| = \sum_{j=1}^n 0 = 0.$$

Hence $\|x\|_1 = 0$. ■

Case $\|x\|_\infty$

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} = \max\{0, \dots, 0\} = 0.$$

Hence $\|x\|_\infty = 0$. ■

Conclusion

Both directions hold for each norm, so $\|x\|_1 = 0 \iff x = 0$ and $\|x\|_\infty = 0 \iff x = 0$. Thus $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are both positive definite. ■